

Problem 2 — SIS CVP Solution Report

Hermes Agent + B&B GSO Pruning

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1 Problem Statement

Find $v, u \in \mathbb{Z}^{100}$ such that $Av + u \equiv t \pmod{100}$ with $\|v\|_\infty \leq 15$, $\|u\|_\infty \leq 15$.

2 Methods Attempted

2.1 BKZ Progressive Reduction

BKZ	Time	Homog. best	Babai CVP
40	3s	score=32	score=45
50	15s	—	—
65	2min	score=21	score=31
70	7min	score=17	score=30
75	1.5h	score=15	score=24
80	3.4h	score=14	score=24
85	15.8h	killed	killed

BKZ-80 produced homogeneous $\ell_\infty = 14$ (3 rows ≤ 15). Babai CVP stuck at 24.

2.2 Complete Method Comparison

Method	Best Score	Verdict
Babai CVP (BKZ-80)	24	baseline
Babai + short homog. rows	21	local minimum
Randomized Babai (50K)	24	no improvement
Simulated annealing	24	random steps destructive
Greedy coordinate descent	22	myopic, stuck
Targeted row combinations	21	wrong subspace
LLL + Kannan (corrected u)	44	far worse than Babai
G6K workout (bs 40–80)	24–26	no CVP gain
fpylll CVP.closest_vector	—	3.8h, no result
Z3 / CP-SAT	—	timeout

3 Successful Solution: B&B GSO Pruning

3.1 Preprocessing

Lattice basis (BKZ-80, 200×200): $B = \begin{bmatrix} I_{100} & -A^T \\ 0 & qI_{100} \end{bmatrix}$ (3h23m, fpylll, strategies=default.json).

Babai CVP gave initial $\|v\|_\infty = 24$, $\|u\|_\infty = 19$.

3.2 Algorithm

GSO vectors $b_1^*, \dots, b_{200}^*$ with $r_j = \|b_j^*\|^2$. Babai coefficients: $c_j = \lfloor \langle t_{\text{curr}}, b_j^* \rangle / r_j \rfloor$ ($j = 199 \rightarrow 0$).

Search: last $K = 8$ coefficients ($j = 192 \dots 199$), each ± 4 around Babai value. Combinations: $9^8 \approx 4.3 \times 10^7$.

Pruning: at level j , ℓ_2 lower bound from dims $0 \dots j-1$: $\ell_2^{\text{lb}} = \sqrt{\sum_{k=0}^{j-1} r_k \cdot \text{frac}(\langle t_{\text{res}}, b_k^* \rangle / r_k)^2}$. Prune if $\ell_2^{\text{lb}} > \ell_2^{\text{best}}$. Dims $0 \dots 191$ use Babai rounding only.

3.3 Result

$$24 \rightarrow 23 \rightarrow 22 \rightarrow 20 \rightarrow 18 \rightarrow 17 \rightarrow 16 \rightarrow \mathbf{15}$$

6 minutes, 43M combinations.

$\|v\|_\infty = 15$, $\|u\|_\infty = 15$, $Av+u \equiv t \pmod{100} \checkmark$ Solution: `results_p2/solution_p2_20260521_220739.txt`.

4 Why B&B Was Orders of Magnitude Faster

4.1 Right Subspace

BKZ-80 GSO norm profile:

- Dims 0–191: r_j large ($\sim 10^4$), Babai rounding is accurate.
- Dims 192–199: r_j small but $\mu_{j,k}$ propagate corrections to all 200 coordinates.

The Babai error lives almost entirely in the last 8 GSO dimensions.

4.2 Right Pruning Metric

ℓ_∞ pruning kills all branches at the root (partial v have large ℓ_∞ from first rows). ℓ_2 bound accurately reflects remaining improvement room on well-reduced bases.

4.3 Head-to-Head

Failed Method	Why B&B Beats It
Simulated annealing	Random perturbation has no structural guidance. B&B explores structured GSO perturbations only.
Greedy descent	Fixes one coordinate at a time, blind to multi-coefficient corrections. B&B searches all 8 simultaneously.
Short homog. rows	Their large-coordinate positions do not overlap Babai's bad coordinates. B&B uses full GSO structure.
fpylll CVP exact	Searches all 200 dims — tree is astronomically large. B&B enumerates only 8.
G6K workout	Improves basis, not CVP. Babai is already locally optimal.
LLL + Kannan	CVP distance ($\ell_2 \approx 212$) $>$ λ_1 ($\ell_2 \approx 198$). The $c = -1$ solution is never shortest.

4.4 Core Insight

The CVP solution is only $24 \rightarrow 15$ away from Babai in ℓ_∞ . This gap is concentrated in the last few GSO dimensions, where small coefficient changes produce large global effects via the μ matrix. Eight dimensions ± 4 with GSO-guided pruning finds it in minutes.

5 Conclusion

1. **BKZ-80** (3.4h) — essential preprocessing on correct SIS basis.
2. **Babai CVP** — baseline (score=24), locally optimal, small perturbations cannot improve.
3. **B&B GSO pruning** (6min) — enumerates last 8 GSO coefficients ± 4 , ℓ_2 pruning, escapes local minimum to score=15.